MAE3405 Aerospace Propulsion

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Topic 9-10 Problem Set Questions

An asterisk indicates that this problem is similar to a past exam question.

Question 20

A single spool turbojet with a fixed overall pressure ratio of 20 and maximum cycle temperature of 1400 K operates with an ideal compressor, turbine and nozzles.
Assume that the specific heats are the same for cold and hot sides.
Assume fuel mass flow is negligible with respect to air mass flow.

(i) For a fixed air speed of Mach=0.8, plot the specific thrust and TSFC against altitude up to 10,000m.

(ii) For a fixed altitude of 5,000m, plot the specific thrust and TSFC for air speeds ranging from Mach=0 to 0.8.

[Ask demonstrator to check graphs. Ans: (i) at 10,000m, Fs=].

Question 21

A turbojet is operating with the conditions below.

Quantity	**Value**	**Quantity**	**Value**
Compressor Ratio	9.0	Compressor Cp	1.005kJ/kg.K
Turbine Inlet Temp	1250K	Compressor γ	1.4
Isentropic Efficiencies		Turbine Cp	1.148kJ/kg.K
Compressor	87%	Turbine γ	1.333
Turbine	87%	Exhaust Gas Constant R	0.287kJ/kg.K
Intake	95%	Engine Mass Flow	15kg/s
Propelling Nozzle	95%	Airspeed	260m/s
Mechanical Efficiency	99%	Altitude	7000m
Combustion Efficiency	98%	Combustor Pressure drop	6%

Calculate the propelling nozzle area required, the net thrust developed and the TSFC (in kg/kN.hr) using the Cohen et al. combustor temperature diagram from the notes. [Ans: 0.0644 m^2; 8.86 kN; 0.120 kg/N.h]

Question 22

The exhaust gases in the jet pipe of the engine above are reheated to 2000K by supply of Jet A1 fuel to an afterburner; the combustion pressure loss incurred is 3% of the pressure at the outlet from the turbine. If the engine intake mass flow rate is to remain unchanged, calculate:

(i) the percentage increase in nozzle area required,
(ii) the percentage increase in net thrust,
(iii) the percentage increase in fuel flow rate & TSFC.

[Ans: 49%; 63%; 144% & 50%]

Question 23*

For the single spool turbo-prop engine below, determine the thrust power assuming the propeller to be 85% efficient and the propelling nozzle pressure ratio to be 1.3. [Ans: 1.86 MW]

Quantity	**Value**	**Quantity**	**Value**
Compressor Ratio	8.0	Combustion Pressure drop	4%
Turbine Inlet Temp	1200K	Compressor Cp	1.005kJ/kg.K
Isentropic Efficiencies		Compressor γ	1.4
-Compressor	87%	Turbine Cp	1.148kJ/kg.K
-Turbine	90%	Turbine γ	1.333
-Intake	93%	Exhaust Gas Constant R	0.287kJ/kg.K
-Propelling Nozzle	95%	Intake Nozzle Area	0.1m^2
Spool Efficiency	99%	Altitude	5000m
Combustion Efficiency	98%	Airspeed	125m/s
Gear Box efficiency	90%	Fuel	Jet-A1

Question 24A

A new turbo-prop engine is being designed. As well as having a gas generator spool, the engine also has a second power spool which drives the propeller through a gear box. As well as having a combustor located between the compressor and turbine of the gas generator spool, the engine has a re-heat combustor between the exit of the gas generator turbine and the power turbine. This is designed to increase thrust power. For the engine specifications below, determine the thrust power for a take-off speed of 75 knots and propulsive efficiency 85% at SSL.
[Ans: 1381 kW]

Quantity	**Value**	**Quantity**	**Value**
Compressor Ratio	10	Compressor Cp	1.005kJ/kg-K
Combustor F/A ratio	0.015	Compressor γ	1.4
Isentropic Efficiencies		Turbine Cp	1.148kJ/kg-K
- Compressor	85%	Turbine	1.333
- Turbine and power turb.	90%	Exhaust gas constant R	0.287kJ/kg-K
- Intake (subsonic)	95%	Intake nozzle area	0.2m^2
- Propelling nozzle	95%	Reheat F/A ratio	0.005
Gas generator spool eff.	99%	Reheat comb. eff	98%
Combustor efficiency	98%	Reheat pressure drop	2%
Combustor pressure drop	3%	Gear box efficiency	85%
Propelling noz. press ratio	1.2	Fuel	Jet A1

Question 25

The following data is for a twin-spool turbofan gas turbine engine, with the fan driven by the LP turbine and the compressor by the HP turbine. Separate cold and hot nozzles are used. Determine the net thrust and TSFC.
[Ans: 29 kN, 74.1 kg/kN-hr]

Quantity	**Value**	**Quantity**	**Value**
Overall Pressure Ratio	20.0	Compressor Cp	1.005kJ/kg.K
Turbine Inlet Temp	1300K	Compressor γ	1.4
Fan Pressure Ratio	1.7	Turbine Cp	1.148kJ/kg.K
Compressors, Fan & Turbine Isentropic efficiency	90%	Turbine γ	1.333
Intake Isentropic Eff.	95%	Exhaust Gas Constant R	0.287kJ/kg.K
Propelling Nozzles Eff.	95%	Airspeed	Mach 0.8
Spool Mechanical Eff.	99%	Altitude	10,000m
Combustion Efficiency	98%	Area Cold Nozzle	0.8m^2
Combustion Pressure drop	4%	By-pass ratio	4

Question 26

For a ramjet, plot the specific thrust and TSFC against Mach number for speeds, from Mach number 0.75 up to 5, for an altitude of 20,000m. Assume that the maximum flame temperature is 2200K and combustion pressure drop is 1% of the ram pressure.

The intake may be considered as being divided into two sections; supersonic and subsonic. In the supersonic section, should the Mach number be greater than 1, use is made of a normal shock in the inlet plane to reduce the airspeed to subsonic conditions. The subsonic component of the inlet leading to the combustor section has an isentropic efficiency of 95% while the converging propelling nozzle has an isentropic efficiency of 95%. Assume the usual air and product property data. Comment on the results.

[Ans: At Ma=1.75; F_s= 752 Ns/kg TSFC= 0.25 kg/N-h.
Ask a demonstrator for other Mach Numbers & check graphs]

Question 27

Repeat question 26 but assume after an airspeed of Mach=1 an isentropic converging- diverging propelling nozzle is used to fully expand the exhaust gas to atmospheric pressure. Given the nozzle area ratio, is acceleration from Mach 1 to Mach 5 using the same nozzle system practical?

[Ans: Ma=1.75 F_s= 791 Ns/kg TSFC= 0.23 kg/N.h
Ask a demonstrator for other Mach Numbers & check graphs]

Question 28B

A ram jet operates on fuel which can by approximated by $C_{10}H_{22}$. The combustor fuel/air ratio is 0.03. The inlet conditions consist of a single oblique shock at a wave angle of 30o terminated by a normal shock, followed by subsonic flow through the entry nozzle into the combustor. The exit conditions are a fully expanded convergent-divergent propelling nozzle. The efficiency of the combustor and nozzles is 95%, and combustor pressure loss is 5% of the ram pressure. For specific heats, assume the fuel is similar to Jet A1. Determine the specific thrust for a cruise speed of Mach 2.5 at an altitude of 20,000m.

[Ans: 613.4 Ns/kg]